



Smart Event & Hall Reservation System

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1. Problem Statement

Many event halls such as wedding halls, church halls, school auditoriums, and community centers still rely on manual registers or phone calls to manage reservations. This often leads to double bookings, scheduling conflicts, loss of customer information, and difficulty in tracking payments and availability.

This project aims to develop a Windows Forms Application using C# and the .NET Framework to efficiently manage hall reservations, customer details, event schedules, and payments through a centralized and user-friendly system.

2. Objectives of the System

- Prevent double bookings of halls.
- Digitally manage reservations and customer records
- Track payments and reservation status
- Generate daily and monthly reservation reports.

3. Team Members and Their Roles

Name	Registration Number	Role
Member 1	2022/ICT/33	Payment Form
Member 2	2022/ICT/64	Dashboard Form
Member 3	2022/ICT/07	Login Form
Member 4	2022/ICT/89	Hall Form
Member 5	2022/ICT/139	Reservation Form
Member 6	2022/ICT/114	Report Form



4. Technologies Used

- **Programming Language:** C#
- **.NET Framework**
- **IDE:** Visual Studio
- **Database:** SQL Server
- **Application Type:** Windows Forms Application



5. System Overview

The system is a Windows Forms Application consisting of multiple interconnected forms. It allows users to input, view, update, and delete records efficiently using a database. The system ensures user-friendly interaction and reliable data management.



6. Description of Forms

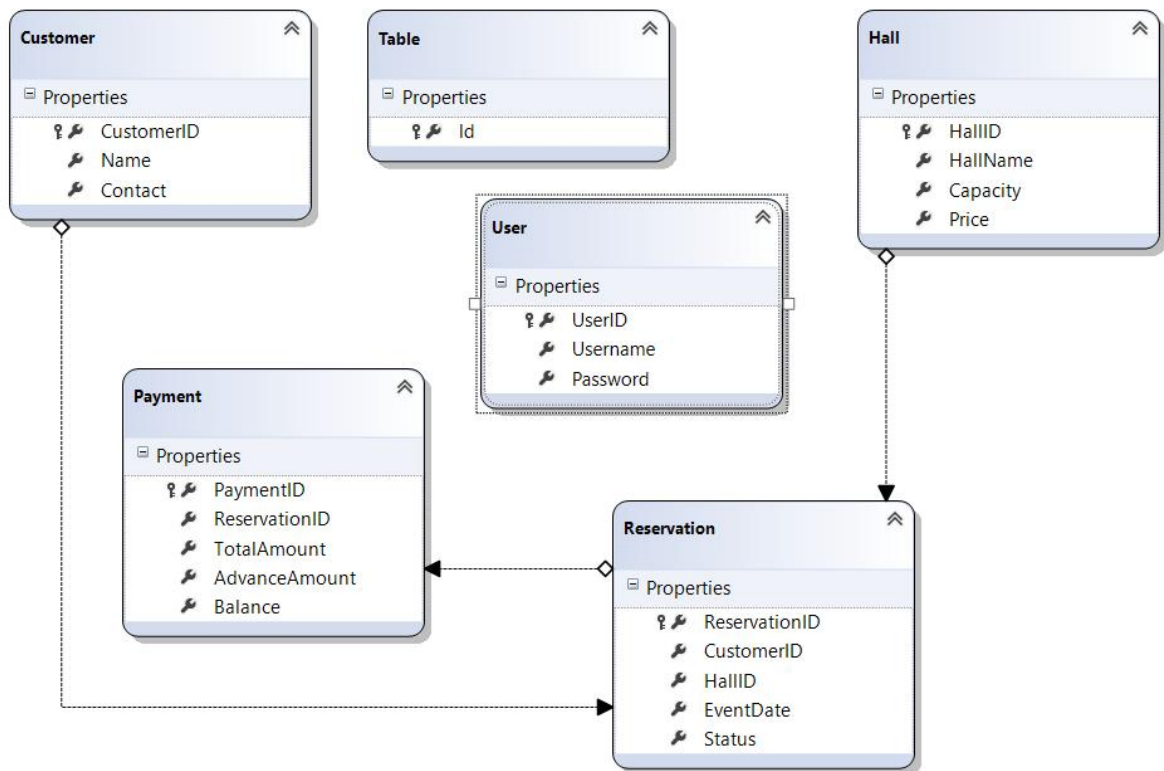
1. **Login Form** – Used to authenticate users and allow secure access to the system.
2. **Dashboard Form** – Acts as the main navigation hub for the applications.
3. **Data Entry Form** – Used to add new records to the database.
4. **View Records Form** – Displays stored data using DataGridView.
5. **Update/Delete Form** – Allows modification and deletion of existing records.
6. **About/Help Form** – Provides information about the application.



7. Application Screenshots

Screenshots of each form are included below with descriptions.

DataClasses



Login Form

```
using System;
using System.Linq;
using System.Windows.Forms;
using System.Text.RegularExpressions;

namespace Group_Project
{
    3 references
    public partial class LoginForm : Form
    {
        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public LoginForm()
        {
            InitializeComponent();

            this.FormBorderStyle = FormBorderStyle.FixedSingle;
            this.MaximizeBox = false;
        }

        1 reference
        private void Login_Click(object sender, EventArgs e)
        {
            try
            {
                var user = db.Users.FirstOrDefault(u => u.Username == textBox1.Text
                && u.Password == textBox3.Text);

                if (user != null)
                {
                    DashboardForm dash = new DashboardForm();
                }
            }
        }
    }
}
```

```
        if (user != null)
        {
            DashboardForm dash = new DashboardForm();
            dash.Show();
            this.Hide();
        }
        else
        {
            MessageBox.Show("Invalid Username or Password!", "Login Error",
            MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}

1 reference
private void button1_Click(object sender, EventArgs e)
{
    string username = textBox1.Text.Trim();
    string password = textBox3.Text.Trim();

    if (string.IsNullOrEmpty(username) || string.IsNullOrEmpty(password))
    {
        MessageBox.Show("Please enter both Username and Password.");
        return;
    }

    Regex nameRegex = new Regex("[A-Z][a-zA-Z]*");
}
```

```
    Regex nameRegex = new Regex("[A-Z][a-zA-Z]*");
    if (!nameRegex.IsMatch(username))
    {
        MessageBox.Show("Username must start with a Capital letter and contain only letters.",
        "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    Regex passwordRegex = new Regex("^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-9])(?=.*[!@#$%^&*~]).{8,}$");
    if (!passwordRegex.IsMatch(password))
    {
        MessageBox.Show("Password is too weak!\n\nMust have:\n- Minimum 8 " +
        "characters\n- One Uppercase letter\n- One Lowercase letter\n- One Number\n- One Special character",
        "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
        return;
    }

    try
    {
        bool userExists = db.Users.Any(u => u.Username == username);

        if (userExists)
        {
            MessageBox.Show("This username already exists. Please choose another one.");
        }
        else
        {
            User newUser = new User
            {
                Username = username,
                Password = password
            };

            db.Users.InsertOnSubmit(newUser);
            db.SubmitChanges();

            MessageBox.Show("Account created successfully! Now you can log in.",
            "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);

            textBox1.Clear();
            textBox3.Clear();
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Signup Error: " + ex.Message);
    }
}

1 reference
private void button2_Click_1(object sender, EventArgs e)
{
    Application.Exit();
}
```

```
        User newUser = new User
        {
            Username = username,
            Password = password
        };

        db.Users.InsertOnSubmit(newUser);
        db.SubmitChanges();

        MessageBox.Show("Account created successfully! Now you can log in.",
        "Success", MessageBoxButtons.OK, MessageBoxIcon.Information);

        textBox1.Clear();
        textBox3.Clear();
    }
    catch (Exception ex)
    {
        MessageBox.Show("Signup Error: " + ex.Message);
    }
}

1 reference
private void button2_Click_1(object sender, EventArgs e)
{
    Application.Exit();
}
```


DashboardForm.

Dashboard

Logout

Total Booking: 1

Available Halls: 3

Upcoming Events: 1

Total Earning: Rs 500

Manage Halls

Reservations

Payments

Reports

Upcoming Events

Show All

EventDate	Hall_Name	Customer_Name
2/3/2026	Saketha	Kavishka

Welcome, Admin!

```
using System;
using System.Data;
using System.Linq;
using System.Windows.Forms;

namespace Group_Project
{
    7 references
    public partial class DashboardForm : Form
    {
        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public DashboardForm()
        {
            InitializeComponent();
        }

        1 reference
        private void DashboardForm_Load(object sender, EventArgs e)
        {
            RefreshDashboard();
        }

        4 references
        public void RefreshDashboard()
        {
            try
            {

```

```
Group_Project
+ Group_Project.DashboardForm
+ DashboardForm()

try
{
    label4.Text = db.Reservations.Count().ToString();
    label6.Text = db.Halls.Count().ToString();
    label2.Text = db.Reservations.Count().ToString();

    decimal income = db.Payments.Sum(p => (decimal?)p.AdvanceAmount ?? 0);
    label8.Text = "Rs." + income.ToString("N0");

    LoadUpcomingEvents();
}
catch (Exception ex)
{
    MessageBox.Show("Error refreshing data: " + ex.Message);
}

2 references
private void LoadUpcomingEvents()
{
    dataGridView1.DataSource = db.Reservations.Select(r => new {
        EventDate = r.EventDate,
        Hall_Name = r.Hall.HallName,

```

```
dataGridView1.DataSource = db.Reservations.Select(r => new {
    EventDate = r.EventDate,
    Hall_Name = r.Hall.HallName,
    Customer_Name = r.Customer.Name
}).ToList();

1 reference
private void button2_Click(object sender, EventArgs e) { new ReservationForm().Show(); }
1 reference
private void button3_Click(object sender, EventArgs e) { new HallForm().Show(); }
1 reference
private void button5_Click(object sender, EventArgs e) { new ReportForm().Show(); }

0 references
private void button6_Click(object sender, EventArgs e)
{
    LoadUpcomingEvents();
}

1 reference
private void label2_Click(object sender, EventArgs e) { }
1 reference
private void label4_Click(object sender, EventArgs e) { }
```

HallForm

HallID	HallName	Capacity	Price
1	Saketha	20	1500
2	Read Rose	120	1500

```
using System;
using System.Linq;
using System.Windows.Forms;

namespace Group_Project
{
    3 references
    public partial class HallForm : Form
    {
        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public HallForm()
        {
            InitializeComponent();
        }

        1 reference
        private void HallForm_Load(object sender, EventArgs e)
        {
            LoadData();
        }

        4 references
        private void LoadData()
        {
            try
            {

```

```
3 references
        private void ClearFields()
        {
            textBox1.Clear();
            textBox2.Clear();
            textBox3.Clear();
            textBox4.Clear();
        }

        5 references
        private void HallForm_FormClosed(object sender, FormClosedEventArgs e)
        {
            foreach (Form f in Application.OpenForms)
            {
                if (f is DashboardForm dash)
                {
                    dash.RefreshDashboard();
                    break;
                }
            }
        }

        1 reference
        private void button1_Click(object sender, EventArgs e)
        {
            if (string.IsNullOrEmpty(textBox1.Text) || string.IsNullOrEmpty(textBox2.Text) || string.IsNullOrEmpty(textBox3.Text) || string.IsNullOrEmpty(textBox4.Text))
            {
                MessageBox.Show("Please fill in all the information.");
                return;
            }
            try
            {
                Hall h = new Hall
                {

```

```
try
            {
                Hall h = new Hall
                {
                    HallName = textBox1.Text.Trim(),
                    Capacity = int.Parse(textBox2.Text),
                    Price = decimal.Parse(textBox3.Text)
                };

                db.Halls.InsertOnSubmit(h);
                db.SubmitChanges();

                MessageBox.Show("Hall added successfully!");
                ClearFields();
                LoadData();
            }
            catch (Exception ex)
            {
                MessageBox.Show("Insertion Error: " + ex.Message);
            }
        }

        1 reference
        private void button2_Click(object sender, EventArgs e)
        {
            try
            {
                if (string.IsNullOrEmpty(textBox4.Text)) { MessageBox.Show("Please enter a Hall ID to search."); return; }

                int id = int.Parse(textBox4.Text);
                var upHall = db.Halls.FirstOrDefault(h => h.HallID == id);

                if (upHall != null)
                {

```

```
if (upHall != null)
        {
            upHall.HallName = textBox1.Text.Trim();
            upHall.Capacity = int.Parse(textBox2.Text);
            upHall.Price = decimal.Parse(textBox3.Text);

            db.SubmitChanges();
            MessageBox.Show("Information updated successfully!");
            LoadData();
        }
        else
        {
            MessageBox.Show("Hall record not found.");
        }
    }
    catch (Exception ex) { MessageBox.Show("Update Error: " + ex.Message); }

    1 reference
    private void button3_Click(object sender, EventArgs e)
    {
        try
        {
            if (string.IsNullOrEmpty(textBox4.Text)) { MessageBox.Show("Please enter a Hall ID to delete."); return; }

            int id = int.Parse(textBox4.Text);
            var delHall = db.Halls.FirstOrDefault(h => h.HallID == id);

            if (delHall != null)
            {
                db.Halls.DeleteOnSubmit(delHall);
                db.SubmitChanges();
                MessageBox.Show("Data deleted successfully!");
                ClearFields();
                LoadData();
            }
        }
    }
}


```

```
else
        {
            MessageBox.Show("Hall record not found.");
        }
    }
    catch (Exception ex) { MessageBox.Show("Delete Error: " + ex.Message); }

    1 reference
    private void button4_Click(object sender, EventArgs e)
    {
        ClearFields();
    }

    1 reference
    private void button5_Click(object sender, EventArgs e)
    {
        try
        {
            if (string.IsNullOrEmpty(textBox4.Text)) { MessageBox.Show("Please enter an " + "ID to search."); return; }

            int id = int.Parse(textBox4.Text);
            var hall = db.Halls.FirstOrDefault(h => h.HallID == id);

            if (hall != null)
            {
                textBox1.Text = hall.HallName;
                textBox2.Text = hall.Capacity.ToString();
                textBox3.Text = hall.Price.ToString();
            }
            else
            {
                MessageBox.Show("No data found for the given ID.");
            }
        }
    }


```

```
1 reference
    private void button5_Click(object sender, EventArgs e)
    {
        try
        {
            if (string.IsNullOrEmpty(textBox4.Text)) { MessageBox.Show("Please enter an " + "ID to search."); return; }

            int id = int.Parse(textBox4.Text);
            var hall = db.Halls.FirstOrDefault(h => h.HallID == id);

            if (hall != null)
            {
                textBox1.Text = hall.HallName;
                textBox2.Text = hall.Capacity.ToString();
                textBox3.Text = hall.Price.ToString();
            }
            else
            {
                MessageBox.Show("No data found for the given ID.");
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show("Search Error: " + ex.Message);
        }
    }

    1 reference
    private void pictureBox1_Click(object sender, EventArgs e) { }


```


PaymentForm

Reservation ID

Total Account

Advance Amount

Balance

```
using System;
using System.Linq;
using System.Windows.Forms;

namespace Group_Project
{
    4 references
    public partial class PaymentForm : Form
    {
        int resID;
        decimal totalPrice;

        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public PaymentForm(int id, decimal price)
        {
            InitializeComponent();
            this.resID = id;
            this.totalPrice = price;
        }

        1 reference
        private void PaymentForm_Load(object sender, EventArgs e)
        {
            textBox1.Text = totalPrice.ToString("F2");

            textBox1.ReadOnly = true;
            textBox3.ReadOnly = true;
        }
    }
}
```

```
private void textBox2_TextChanged(object sender, EventArgs e)
{
    if (decimal.TryParse(textBox2.Text, out decimal advance))
    {
        decimal balance = totalPrice - advance;
        textBox3.Text = balance.ToString("F2");
    }
    else
    {
        textBox3.Text = totalPrice.ToString("F2");
    }
}

1 reference
private void button1_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(textBox2.Text))
    {
        MessageBox.Show("Please enter the Advance Amount.");
        return;
    }

    try
    {
        decimal advance = decimal.Parse(textBox2.Text);
        decimal balance = totalPrice - advance;

        Payment p = new Payment
        {
            ReservationID = resID,
            TotalAmount = totalPrice,
            AdvanceAmount = advance
        }
    }
}
```

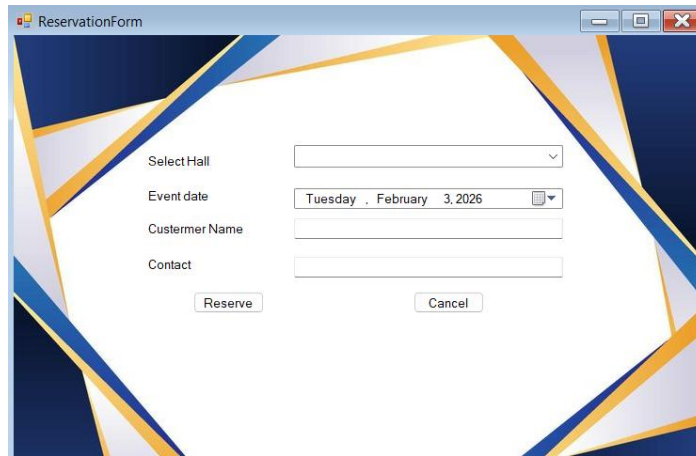
```
Payment p = new Payment
{
    ReservationID = resID,
    TotalAmount = totalPrice,
    AdvanceAmount = advance,
    Balance = balance
};

db.Payments.InsertOnSubmit(p);
db.SubmitChanges();

MessageBox.Show("Payment Successful! Balance: Rs." + balance.ToString("N2"), "Success");
this.Close();
}
catch (Exception ex)
{
    MessageBox.Show("Error processing payment: " + ex.Message);
}
}

1 reference
private void PaymentForm_FormClosed(object sender, FormClosedEventArgs e)
{
    foreach (Form f in Application.OpenForms)
    {
        if (f is DashboardForm dash)
        {
            dash.RefreshDashboard();
            break;
        }
    }
}
```

ReservationForm



```
using System;
using System.Linq;
using System.Windows.Forms;

namespace Group_Project
{
    3 references
    public partial class ReservationForm : Form
    {
        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public ReservationForm()
        {
            InitializeComponent();
        }

        1 reference
        private void ReservationForm_Load(object sender, EventArgs e)
        {
            try
            {
                comboBox1.DataSource = db.Halls.Select(h => h.HallName).ToList();
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error loading halls: " + ex.Message);
            }
        }

        1 reference
        private void button1_Click(object sender, EventArgs e)
        {
            if (string.IsNullOrEmpty(textBox1.Text) || comboBox1.SelectedIndex == -1)
            {
                MessageBox.Show("Please enter Customer Name and select a Hall.");
                return;
            }
        }
    }
}
```

```
try
{
    var hall = db.Halls.First(h => h.HallName == comboBox1.Text);

    Customer c = new Customer
    {
        Name = textBox1.Text.Trim(),
        Contact = textBox2.Text.Trim()
    };
    db.Customers.InsertOnSubmit(c);
    db.SubmitChanges();

    Reservation res = new Reservation
    {
        CustomerID = c.CustomerID,
        HallID = hall.HallID,
        EventDate = dateTimePicker1.Value.Date,
        Status = "Confirmed"
    };
    db.Reservations.InsertOnSubmit(res);
    db.SubmitChanges();

    MessageBox.Show("Reservation saved successfully! Redirecting to Payment...");

    PaymentForm payForm = new PaymentForm(res.ReservationID, hall.Price);
    payForm.FormClosed += (s, args) => RefreshDashboardData();
    payForm.Show();

    this.Hide();
}
catch (Exception ex)
{
    MessageBox.Show("Error processing reservation: " + ex.Message);
}
}
```

```
    }
}

2 references
private void RefreshDashboardData()
{
    foreach (Form f in Application.OpenForms)
    {
        if (f is DashboardForm dash)
        {
            dash.RefreshDashboard();
            break;
        }
    }
}

0 references
private void ReservationForm_FormClosed(object sender, FormClosedEventArgs e)
{
    RefreshDashboardData();
}

1 reference
private void label4_Click(object sender, EventArgs e) { }
```

ReportForm

ReservationID	Date	Hall	Customer	Contact
1	2/3/2026	Saketha	Kavishka	07677990

```
using System;
using System.Data;
using System.Linq;
using System.Windows.Forms;

namespace Group_Project
{
    3 references
    public partial class ReportForm : Form
    {
        DataClasses1DataContext db = new DataClasses1DataContext();

        1 reference
        public ReportForm()
        {
            InitializeComponent();
        }

        0 references
        private void ReportForm_Load(object sender, EventArgs e)
        {
            LoadFullReport();
        }

        2 references
        private void LoadFullReport()
        {
            try
            {
                var report = from r in db.Reservations
                    join h in db.Halls on r.HallID equals h.HallID
                    join c in db.Customers on r.CustomerID equals c.CustomerID
                    orderby r.EventDate descending
                    select new
                    {
                        ReservationID = r.ReservationID,
                        Date = r.EventDate,

```

```

                        ReservationID = r.ReservationID,
                        Date = r.EventDate,
                        Hall = h.HallName,
                        Customer = c.Name,
                        Contact = c.Contact,
                        Status = r.Status
                    };

                dataGridView1.DataSource = report.ToList();
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error loading report: " + ex.Message);
            }
        }

        1 reference
        private void button2_Click(object sender, EventArgs e)
        {
            try
            {
                DateTime selectedDate = dateTimePicker1.Value.Date;

                var filteredReport = from r in db.Reservations
                    join h in db.Halls on r.HallID equals h.HallID
                    join c in db.Customers on r.CustomerID equals c.CustomerID
                    where r.EventDate == selectedDate
                    select new
                    {
                        ReservationID = r.ReservationID,
                        Date = r.EventDate,
                        Hall = h.HallName,
                        Customer = c.Name,
                        Contact = c.Contact,
                        Status = r.Status
                    };

```

```

                    };
                dataGridView1.DataSource = filteredReport.ToList();

                if (!filteredReport.Any())
                {
                    MessageBox.Show("No reservations found for the selected date.");
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show("Search Error: " + ex.Message);
            }
        }

        0 references
        private void button1_Click(object sender, EventArgs e)
        {
            LoadFullReport();
        }

        1 reference
        private void panel2_Paint(object sender, PaintEventArgs e)
        {
        }
    }
}

```



8. Database Design

The application uses SQL Server as the backend database. Tables are designed with primary keys to store user and system data securely. CRUD operations are implemented to manage data effectively.



9. Events and Controls Used

- Button Click Events
- Form Load Event
- TextBox TextChanged Event
- ComboBox SelectionChanged Event
- MessageBox for alerts and confirmations



10. Error Handling and Validation

Proper exception handling is implemented using try-catch blocks. Input validation ensures that users enter correct data, and error messages are displayed using MessageBox controls.



11. Challenges Faced and Solutions

Challenge: Database connection issues.

Solution: Corrected connection string and added exception handling.

Challenge: Form navigation errors.

Solution: Used proper event handling and form references.



12. Conclusion

The project successfully demonstrates the use of C# and Windows Forms to develop a real-world application. Through this project, we gained practical experience in event-driven programming, database connectivity, and team collaboration.